

Project

Numerical Methods for Economics and Finance

Andre Silva, Nova SBE 2025/2026

Pietro De Bernardi (69791) – Marco Aldrighetti (72475) – Alessandro De Riccardis (72086)

Question 1 – Stochastic Consumption and Investment Model

The economy is populated by a representative agent who holds capital k_t , observes productivity A_t , and produces $A_t k_t^\alpha$ goods that are split between consumption c_t and next period capital k_{t+1} . The dynamic problem is

$$v(k_t, A_t) = \max_{k_{t+1}} \{ \log(A_t k_t^\alpha - k_{t+1}) + \beta \mathbb{E}_t v(k_{t+1}, A_{t+1}) \}, \quad (1)$$

with the productivity process

$$A_{t+1} = (1 - \rho) + \rho A_t + \varepsilon_{t+1}, \quad \varepsilon_{t+1} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_\varepsilon^2), \quad (2)$$

and parameters $0 < \alpha < 1$, $0 < \beta < 1$, $0 < \rho < 1$. The long run mean of A_t is $\mu_A = 1$. The numerical analysis uses $\beta = 0.8$, $\alpha = 0.5$, $\rho = 0.95$ and $\sigma_\varepsilon = 0.005$.

(a) Nature of the Problem, Intertemporal Trade, States and Controls

The agent must split current resources $y_t = A_t k_t^\alpha$ between consumption today and investment for tomorrow. Consuming more now raises flow utility $\log c_t$ but reduces the capital stock available next period, which lowers expected future production and therefore future consumption. This is the classical intertemporal trade between current and future consumption, mediated by the discount factor β and by the marginal product of capital $\alpha A_{t+1} k_{t+1}^{\alpha-1}$.

The information structure at time t is summarised by two variables:

- **States** (predetermined or exogenous): the capital stock carried over from the previous period, k_t , and the current level of productivity, A_t . The pair (k_t, A_t) fully characterises the agent's environment at the moment the decision is taken.
- **Control**: the amount of capital to save for the next period, k_{t+1} . Once k_{t+1} is chosen, consumption is residual: $c_t = A_t k_t^\alpha - k_{t+1}$.

The productivity shock ε_{t+1} is the source of uncertainty; expectations $\mathbb{E}_t$ are taken with respect to its distribution.

(b) First Order Condition

Differentiating the right hand side of (1) with respect to k_{t+1} and using the envelope condition $v_k(k_t, A_t) = \alpha A_t k_t^{\alpha-1} / c_t$ yields the Euler equation

$$\frac{1}{c_t} = \beta \mathbb{E}_t \left[\frac{\alpha A_{t+1} k_{t+1}^{\alpha-1}}{c_{t+1}} \right], \quad (3)$$

with $c_t = A_t k_t^\alpha - k_{t+1}$ and $c_{t+1} = A_{t+1} k_{t+1}^\alpha - k_{t+2}$. Equation (3) states that the marginal utility cost of saving one extra unit today equals the discounted expected marginal utility of the additional consumption it produces tomorrow, valued at the gross return $\alpha A_{t+1} k_{t+1}^{\alpha-1}$ (the marginal product of capital).

Since the project asks for an equation in terms of capital at different time periods, we can substitute consumption out explicitly. Using

$$c_t = A_t k_t^\alpha - k_{t+1}, \quad c_{t+1} = A_{t+1} k_{t+1}^\alpha - k_{t+2},$$

the Euler equation can be written as

$$\frac{1}{A_t k_t^\alpha - k_{t+1}} = \beta \mathbb{E}_t \left[\frac{\alpha A_{t+1} k_{t+1}^{\alpha-1}}{A_{t+1} k_{t+1}^\alpha - k_{t+2}} \right]. \quad (4)$$

This is the capital Euler equation implied by the optimal consumption-investment decision.

(c) Dynare Solution

The model is solved with the following mod file. Following the hint, the timing of k_t and A_t is shifted so that Dynare treats k as a predetermined (state) variable: in the mod file k appears with the lag $k(-1)$ in the resource constraint, and A appears similarly with $A(-1)$ in the AR(1) law of motion.

Listing 1: ramsey.mod – stochastic consumption-investment model

```

1 // ramsey.mod -- Question 1, Project
2 // Stochastic consumption-investment model (no depreciation)
3
4 var k, A, c;
5 varexo epst;
6 parameters beta, alpha, rho, sd_e;
7
8 beta = 0.8;
9 alpha = 0.5;
10 rho = 0.95;
11 sd_e = 0.005;
12
13 model;
14 // Resource constraint: c_t = A_t k_{t-1}^alpha - k_t
15 c = A * k(-1)^alpha - k;
16 // Euler equation: 1/c_t = beta * E_t [ alpha A_{t+1} k_t^(alpha
17 // -1) / c_{t+1} ]
18 1/c = beta * ( alpha * A(+1) * k^(alpha-1) / c(+1) );
19 // AR(1) productivity
20 A = (1 - rho) + rho*A(-1) + epst;
21 end;
22
23 initval;
24 k = 0.16;
25 A = 1;
26 c = 0.24;
27 end;
28
29 steady;
30 check;
```

```

31 shocks;
32     var epst;   stderr sd_e;
33 end;
34
35 stoch_simul(irf=40, order=1);

```

Dynare's default output reports the steady state, the policy function and the impulse response functions to a one standard deviation shock. The values returned are reproduced below.

```

>> dynare ramsey.mod

STEADY-STATE RESULTS:
  k      0.160000
  A      1.000000
  c      0.240000

POLICY AND TRANSITION FUNCTIONS:
               k      A
Constant      0.160000  1.000000
k(-1)         0.500000  0.000000
A(-1)         0.152000  0.950000
epst          0.160000  1.000000

```

Figure 1: Dynare output – steady state and policy and transition functions.

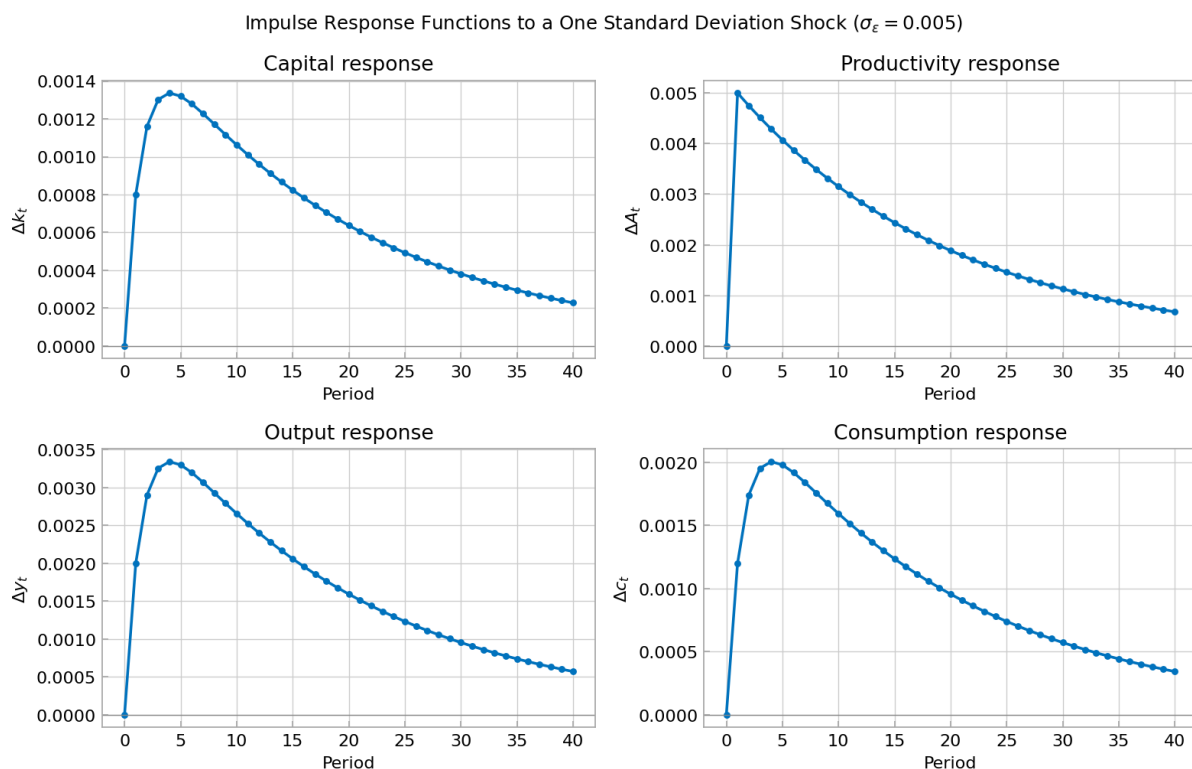


Figure 2: Default Dynare impulse response functions to a one standard deviation productivity shock ($\varepsilon_1 = \sigma_\varepsilon = 0.005$).

(d) Capital Response to a One Unit Productivity Shock

Dynare's default impulse response functions are scaled to one standard deviation innovations. Since $\sigma_\varepsilon = 0.005$, the default Dynare IRF corresponds to a shock of size 0.005, not to a literal

shock of size 1. Because Dynare solves the model using a first-order linear approximation, the response to a one-unit shock can be obtained by scaling the one standard deviation response by

$$\frac{1}{\sigma_\varepsilon} = \frac{1}{0.005} = 200.$$

This preserves the sign and the linear dynamics, but should be interpreted as a linearised response rather than as a fully nonlinear global response to a very large shock.

When the innovation ε_t increases by one unit at $t = 1$ starting from the steady state, productivity jumps from $A_0 = 1$ to $A_1 = 1 + 1 = 2$ and then decays geometrically at rate $\rho = 0.95$. Capital, which is predetermined in period 1 (chosen at the end of period 0), cannot respond contemporaneously; the IRF therefore starts at zero in period 0 and at $\Delta k_1 = a_3 \cdot 1 = 0.16$ in period 1.

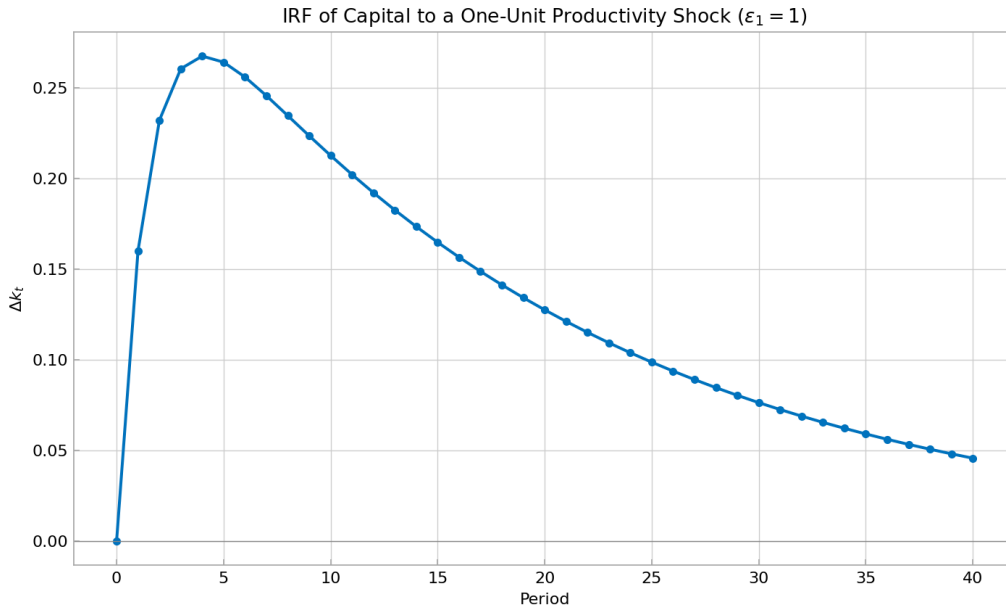


Figure 3: Impulse response of capital to a one-unit productivity shock ($\varepsilon_1 = 1$), obtained by scaling the first-order Dynare response by $1/\sigma_\varepsilon = 200$.

The IRF in Figure 3 reveals a hump shaped path. The capital stock rises sharply in the first few periods, peaks around $t = 4$ at roughly $\Delta k \approx 0.27$ (which corresponds to an increase of about 170% relative to $k_{ss} = 0.16$), and then decays back monotonically to zero. The mechanism behind this shape is the joint dynamics of two competing forces: (i) productivity itself decays at rate $\rho = 0.95$, pulling the marginal product of capital down and reducing the incentive to save; (ii) the previously accumulated capital pushes future output and therefore future investment up, with persistence $a_1 = 0.5$. In the early periods (ii) dominates because the productivity bonus is still large and the capital stock is climbing from below its target; once productivity has decayed enough, (i) takes over and the stock converges back to steady state. The deterministic system is stable because the eigenvalue $a_1 = 0.5$ governing the capital recursion lies strictly inside the unit circle.

(e) Linearised Policy Function Coefficients

Dynare reports the linearised decision rule for capital in the form

$$k_t = a_0 + a_1(k_{t-1} - k_{ss}) + a_2(A_{t-1} - \mu_A) + a_3(\varepsilon_t - \mu_\varepsilon). \quad (5)$$

The four coefficients are obtained by linearising the Euler equation (3) together with the AR(1) law (2) around the deterministic steady state.

Steady state (a_0). Setting $k_{t+1} = k_t = k_{ss}$, $A_t = 1$ and $\varepsilon_t = 0$ in (3) gives $1 = \beta \alpha k_{ss}^{\alpha-1}$, so that

$$k_{ss} = (\beta \alpha)^{\frac{1}{1-\alpha}} = (0.4)^2 = 0.16, \quad c_{ss} = k_{ss}^\alpha - k_{ss} = 0.24, \quad (6)$$

hence $a_0 = k_{ss} = 0.16$.

Stable eigenvalue (a_1). Linearising in logs the Euler equation around $(k_{ss}, 1)$ and imposing the policy form $k_{t+1} = k_{ss} + g_k(k_t - k_{ss}) + g_A(A_t - 1)$, the coefficient on $k_t - k_{ss}$ delivers a quadratic equation in g_k with two roots $\{0.5, 2.5\}$. Blanchard-Kahn selects the stable root inside the unit circle, so $g_k = 0.5 = \alpha$, and after the timing shift this becomes $a_1 = g_k = \alpha = 0.5$.

Productivity loading (a_2 and a_3). The coefficient on $A_t - 1$ in the policy before the timing shift is

$$g_A = -\frac{a_{A_t} + a_{A_{t+1}}\rho}{a_{k_{t+1}} + a_{k_{t+2}}(g_k + \rho)} = k_{ss} = 0.16,$$

where the partials are evaluated at steady state and the algebra simplifies because of the homogeneity of the production function. After Dynare's timing shift the policy reads $k_t = k_{ss} + a_1(k_{t-1} - k_{ss}) + g_A(A_t - 1)$. Substituting $A_t - 1 = \rho(A_{t-1} - 1) + \varepsilon_t$, the contemporaneous innovation and the lagged level of A separate:

$$k_t = k_{ss} + a_1(k_{t-1} - k_{ss}) + g_A\rho(A_{t-1} - 1) + g_A(\varepsilon_t - 0), \quad (7)$$

hence $a_2 = g_A\rho = 0.16 \times 0.95 = 0.152$ and $a_3 = g_A = 0.16 = k_{ss}$.

```
>> ramsey_coefficients

Linearised policy function for capital:
  k_t = a0 + a1*(k_{t-1} - kss) + a2*(A_{t-1} - mu_A)
        + a3*(eps_t - mu_eps)

a0 = 0.160000    (= kss)
a1 = 0.500000
a2 = 0.152000
a3 = 0.160000    (= a2 / rho)
```

Figure 4: Numerical values of the four coefficients in the linearised policy function (5).

The values are

$$a_0 = 0.16, \quad a_1 = 0.50, \quad a_2 = 0.152, \quad a_3 = 0.16, \quad (8)$$

all admitting a clean economic interpretation. The constant a_0 recovers the steady state level. The own coefficient $a_1 = \alpha$ measures how strongly the capital stock self propagates: with $\alpha = 0.5$ half of any deviation from steady state survives to the next period. The two productivity loadings, a_2 and a_3 , both equal k_{ss} up to the AR(1) factor: an exogenous one unit increase in A raises the optimal capital stock by k_{ss} , mirroring the fact that with logarithmic preferences and no depreciation the savings rate $\beta\alpha$ applied to the productivity windfall is exactly the steady state level of capital.

Question 2 – Extended Model with Investment and Depreciation

The economy is the same as in Question 1, but the resource and capital accumulation constraints are now stated separately, and capital depreciates at a constant rate δ :

$$c_t + x_t = y_t, \quad (9)$$

$$y_t = A_t k_t^\alpha, \quad (10)$$

$$k_{t+1} = x_t + (1 - \delta) k_t, \quad (11)$$

$$A_{t+1} = (1 - \rho) + \rho A_t + \varepsilon_{t+1}. \quad (12)$$

where x_t denotes gross investment and $\delta \in (0, 1)$ is the depreciation rate. The remaining parameters $\beta = 0.8$, $\alpha = 0.5$, $\rho = 0.95$ and $\sigma_\varepsilon = 0.005$ are kept as in Question 1. The project introduces depreciation but does not specify a numerical value for δ . The numerical solution uses

$$\delta = 0.10.$$

This value is kept fixed throughout the Dynare solution and the 500-period simulation.

(a) First Order Condition with Depreciation

Substituting (9)-(11) into the Bellman equation collapses the choice to k_{t+1} alone:

$$v(k_t, A_t) = \max_{k_{t+1}} \{ \log(A_t k_t^\alpha + (1 - \delta)k_t - k_{t+1}) + \beta \mathbb{E}_t v(k_{t+1}, A_{t+1}) \}. \quad (13)$$

Differentiating with respect to k_{t+1} yields the FOC

$$\frac{1}{c_t} = \beta \mathbb{E}_t v_k(k_{t+1}, A_{t+1}), \quad (14)$$

and the envelope condition reads

$$v_k(k_t, A_t) = \frac{\alpha A_t k_t^{\alpha-1} + (1 - \delta)}{c_t}, \quad (15)$$

which, compared to Question 1, gains the extra term $(1 - \delta)$ because undepreciated capital carries over with a one to one return. Shifting forward one period and substituting into (14) delivers the modified Euler equation

$$\frac{1}{c_t} = \beta \mathbb{E}_t \left[\frac{\alpha A_{t+1} k_{t+1}^{\alpha-1} + (1 - \delta)}{c_{t+1}} \right]. \quad (16)$$

The economic content is unchanged: the marginal utility cost of saving today equals the discounted expected marginal utility benefit of consuming tomorrow, where each unit of capital now delivers the marginal product plus the undepreciated residual.

(b) Dynare Solution and Impulse Responses

The model is implemented in the mod file below, with c, x, y declared as endogenous variables alongside k and A . The timing convention is the same as in Question 1: k is predetermined and enters as $k(-1)$.

Listing 2: ramsey_dep.mod – extended model with depreciation

```

1 // ramsey_dep.mod -- Question 2, Project
2 // Consumption-investment model with depreciation delta
3
4 var      k, A, c, x, y;
5 varexo   epst;
6 parameters beta, alpha, rho, sd_e, delta;
7
8 beta = 0.8;
9 alpha = 0.5;
10 rho = 0.95;
11 sd_e = 0.005;
12 delta = 0.10;
13
14 model;
15 // Production: y_t = A_t k_{t-1}^alpha
16 y = A * k(-1)^alpha;
17 // Capital accumulation: k_t = x_t + (1-delta) k_{t-1}
18 k = x + (1 - delta) * k(-1);
19 // Resource constraint: c_t + x_t = y_t
20 c + x = y;
21 // Modified Euler equation:
22 // 1/c_t = beta * E_t [ (alpha A_{t+1} k_t^(alpha-1) + (1-delta)
23 // ) / c_{t+1} ]
24 1/c = beta * ( ( alpha * A(+1) * k^(alpha-1) + (1 - delta) ) / c
25 (+1) );
26 // AR(1) productivity
27 A = (1 - rho) + rho * A(-1) + epst;
28 end;
29
30 initval;
31 k = 2.040816;
32 A = 1;
33 y = 1.428571;
34 x = 0.204082;
35 c = 1.224490;
36 end;
37
38 steady;
39 check;
40
41 shocks;
42 var epst; stderr sd_e;
43 end;
44
45 stoch_simul(irf=40, order=1);

```

The steady state values follow from the modified Euler at $A_{ss} = 1$, k_{ss} constant:

$$1 = \beta [\alpha k_{ss}^{\alpha-1} + (1 - \delta)] \implies k_{ss} = \left(\frac{\alpha}{1/\beta - (1 - \delta)} \right)^{\frac{1}{1-\alpha}} \approx 2.0408, \quad (17)$$

with $y_{ss} = k_{ss}^{0.5} \approx 1.4286$, $x_{ss} = \delta k_{ss} \approx 0.2041$ (investment replaces depreciated capital), and $c_{ss} = y_{ss} - x_{ss} \approx 1.2245$. The reported policy and transition values are shown in Figure 5.

```

>> dynare ramsey_dep.mod

STEADY-STATE RESULTS:
k      2.040816
A      1.000000
y      1.428571
x      0.204082
c      1.224490

POLICY AND TRANSITION FUNCTIONS (k row only):
Constant  2.040816
k(-1)     0.832502
A(-1)     0.684111
epst      0.720116

```

Figure 5: Dynare output – steady state and linearised policy function for the extended model.

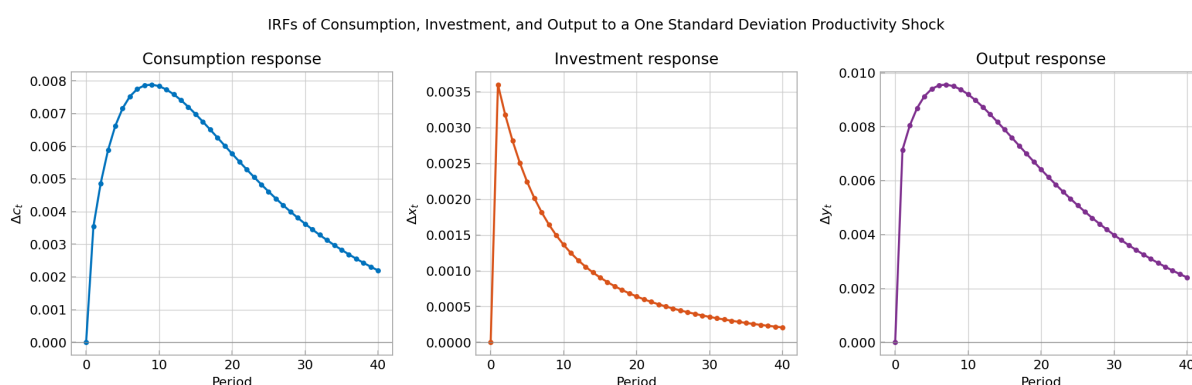


Figure 6: Impulse response functions of consumption c_t , investment x_t , and output y_t to a one standard deviation positive productivity shock.

The three IRFs in Figure 6 display the textbook real business cycle pattern. A positive productivity innovation $\varepsilon_1 = \sigma_\varepsilon$ raises A_1 contemporaneously, but capital k_0 at the start of period 1 is still at steady state, so output $y_1 = A_1 k_0^\alpha$ rises by the full productivity windfall: $\Delta y_1 / y_{ss} \approx 0.50\%$. The household optimally splits this windfall between consumption and investment. Consumption increases by about 0.29% on impact, while investment jumps by 1.76% relative to its (small) steady state value x_{ss} . The asymmetry reflects two forces.

First, the agent wishes to smooth consumption over time: with a log utility and a persistent shock ($\rho = 0.95$) the optimal path raises consumption only gradually, peaking at a 0.64% gain around $t = 9$, and reverting slowly thereafter. Second, the temporarily higher marginal product of capital encourages a sharp burst of investment to build the larger capital stock that supports future production; once the new stock is in place, the investment incentive fades and x_t decays monotonically back to x_{ss} . Output peaks around $t = 7$ at roughly 0.96% above steady state, with the delay reflecting the time it takes for the additional capital to feed back into production.

(c) Simulation over 500 Periods

Setting `periods=500` inside `stoch_simul` instructs Dynare to simulate the model for 500 periods. The simulated paths are recovered from `oo_.endo_simul` (one row per declared variable, 500 columns). The three macroeconomic aggregates c_t , x_t and y_t are plotted in Figure 7.

Listing 3: `ramsey_dep_sim.mod` – 500-period simulation block


```

1 // ramsey_dep_sim.mod -- Question 2(c), Project
2 // Same as ramsey_dep.mod but with periods=500 inside stoch_simul
3
4 // ... (declarations, parameters, model, initval, steady, check,
5 //       shocks
6 //       blocks identical to ramsey_dep.mod) ...
7
8 stoch_simul(periods=500, irf=40, order=1);
9
10 // Plot the simulated paths
11 figure;
12 subplot(3,1,1);
13 plot(oo_.endo_simul(3,:), 'b-', 'LineWidth', 0.9); hold on;
14 yline(1.224490, 'k--', 'LineWidth', 0.9);
15 xlabel('Period t'); ylabel('c_t'); title('Consumption'); grid on;
16
17 subplot(3,1,2);
18 plot(oo_.endo_simul(4,:), 'r-', 'LineWidth', 0.9); hold on;
19 yline(0.204082, 'k--', 'LineWidth', 0.9);
20 xlabel('Period t'); ylabel('x_t'); title('Investment'); grid on;
21
22 subplot(3,1,3);
23 plot(oo_.endo_simul(5,:), 'm-', 'LineWidth', 0.9); hold on;
24 yline(1.428571, 'k--', 'LineWidth', 0.9);
25 xlabel('Period t'); ylabel('y_t'); title('Output'); grid on;

```

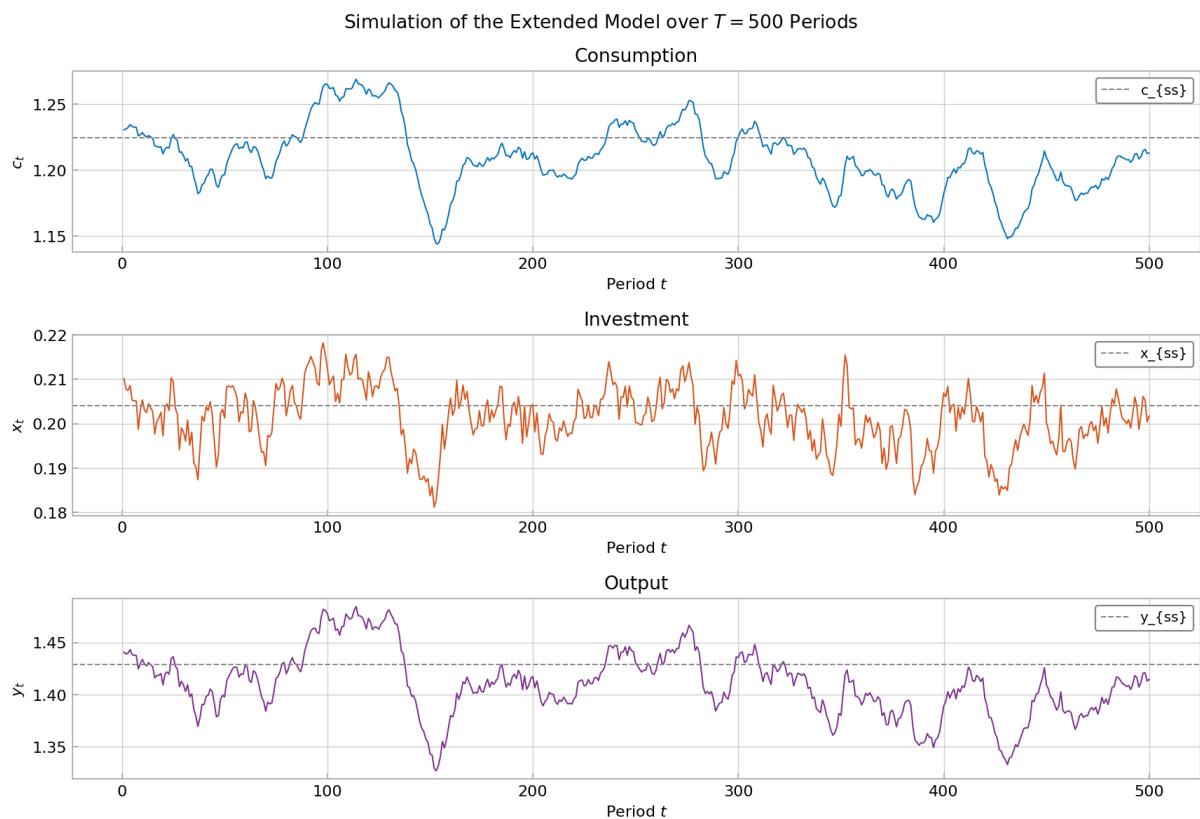


Figure 7: Simulated paths of consumption, investment and output over $T = 500$ periods (dashed line = deterministic steady state).

```

>> size(oo_endo_simul)

ans =      5      500

>> summary_stats

Sample statistics over T = 500 periods:
mean(c) = 1.2089      std(c) = 0.025709
mean(x) = 0.2014      std(x) = 0.006788
mean(y) = 1.4103      std(y) = 0.031110

Relative volatility:  std(x)/std(c) = 0.2640
                     std(c)/std(y) = 0.8264

```

Figure 8: Sample statistics for the 500 period simulation.

The three series are visibly comoving, an immediate consequence of the single source of fluctuation (the productivity shock) and the linear structure of the decision rule. The sample standard deviations, expressed as percentages of the respective steady-state values, are approximately 2.10% for consumption, 3.33% for investment, and 2.18% for output. Therefore, investment is the most volatile aggregate in relative terms.

In absolute terms, $\sigma(x) < \sigma(c)$, because the steady-state level of investment is much smaller than the steady-state level of consumption: $x_{ss} \ll c_{ss}$. In relative terms, however, investment is more volatile. Using standard deviations as percentages of steady-state values, relative investment volatility is approximately

$$\frac{3.33}{2.10} \approx 1.59$$

times consumption volatility. Consumption volatility relative to output volatility is approximately

$$\frac{2.10}{2.18} \approx 0.96.$$

This reproduces the standard real business cycle ranking in relative terms: investment is the most volatile aggregate, while consumption is smoother than investment. The mechanism is the same as in part (b): when a positive shock hits, the household absorbs part of the immediate output gain through investment to build the optimal capital stock, leaving consumption to adjust more gradually through the smoothing motive embedded in the Euler equation (16).